Abhishek Rout

Curriculum Vitae · October 2026

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Research Interests

Modified gravity, in particular Chern–Simons and other parity-violating extensions of general relativity; black hole and cosmological perturbation theory; gravitational-wave phenomenology; quantum field theory in curved spacetime; early-universe cosmology.

Education

Ph.D. in Physics, University of South Carolina, Columbia, SC, USA 2019–2025

Awarded 15 December 2025 · GPA 3.82/4.00
Advisor: Prof. Brett Altschul
Dissertation: *Gravitational Perturbations and Quantum Field Interactions in Modified Spacetime Backgrounds*

Integrated BS–MS in Physics, IISER Kolkata, India 2014–2019

CGPA 8.74/10 (top 10% of cohort)

Publications

Peer-Reviewed

1. A. Rout and B. Altschul, "First-Order Axial Perturbation of the Reissner–Nordström Metric Against a Possible Parity-Violating Gravity Background," *Symmetry* **18**, 560 (2026), doi:10.3390/sym18040560, arXiv:2510.01106 [gr-qc].
2. A. Rout and B. Altschul, "Primordial Gravitational Wave Birefringence in a de Sitter Background with Chern–Simons Coupling," *Universe* **12**, 6 (2026), doi:10.3390/universe12010006, arXiv:2512.03919 [gr-qc].
3. A. Rout and B. Altschul, "Bound States and Particle Production by Breather-Type Background Field Configurations," *Symmetry* **16**, 1571 (2024), doi:10.3390/sym16121571, arXiv:2410.09273 [hep-th].

Submitted

4. A. Rout and B. Altschul, "Gravitational Wave Parity-Violating Strain from Pulsar Glitches in Chern–Simons Modified Gravity," submitted to *Phys. Rev. D*, arXiv:2608.24002[gr-qc].
5. A. Rout, "Analytic parity-violating greybody factors of a dynamical Chern–Simons black hole," submitted to *General Relativity and Gravitation*, arXiv:2610.10280[gr-qc]
6. A. Rout, "Isospectrality Breaking in the Axial Quasinormal Modes of Reissner–Nordström Black Holes in Dynamical Chern–Simons Gravity," submitted to *Phys. Rev. D*, arXiv:2610.38742[gr-qc]
7. A. Rout and B. Altschul, "Mixed-Order Parity-Violating Corrections to Rotating Black Holes in Combined Scalar–Gauss–Bonnet and Dynamical Chern–Simons Gravity," submitted to *Phys. Rev. D*, arXiv:2611.04945[gr-qc]

In Preparation

8. A. Rout and B. Altschul, "Rotation-Induced Isospectrality Breaking of Charged Black Holes in Dynamical Chern–Simons Gravity."
9. A. Rout, "Quasinormal-mode-driven pseudoscalar radiation in dynamical Chern–Simons ringdown."
10. A. Rout and B. Altschul, "Parity-violating spherical photon and timelike orbits in dynamical Chern–Simons gravity: a non-degenerate, sector-separating discriminator from linear to quadratic order in spin."

Research Experience

Doctoral Research, University of South Carolina 2019–2025

Advisor: Prof. Brett Altschul. Parity-violating extensions of general relativity, principally Chern–Simons

modified gravity: axial perturbations of Reissner–Nordström black holes; chiral corrections to tensor modes in de Sitter space and the resulting primordial gravitational-wave birefringence; and parity-violating gravitational-wave strain from pulsar glitches. A separate project treated fermion bound states and particle production in oscillating, breather-type background fields. The work is largely analytical, supported by numerical and symbolic computation in Python and Mathematica.

Master's Thesis, IISER Kolkata 2018–2019

Supervisor: Prof. Narayan Banerjee. Thesis: Gravitational Collapse in a Spherically Symmetric Space-time in presence of a Scalar Field.
Analytical study of gravitational collapse in spherically symmetric spacetimes, including scalar-field-driven collapse with quadratic potentials in dust-dominated cosmological models.

Undergraduate Research Project, IIST Trivandrum 2017

Supervisor: Prof. Sudheesh Chhetil. Loop quantum gravity and spin-network states; applications of entangled loop quantum states to quantum-optics protocols.

Advanced Theoretical Project, IISER Kolkata 2016

Numerical study of the twin paradox in Schwarzschild spacetime (Python).

Talks and Workshops

- "Chern-Simons theory in Reisnner-Nördstrom background", contributed talk, IUCSS Workshop on Theoretical Aspects of the Standard-Model Extension and Modified Gravity, Indiana University, Bloomington, July 2024.
- "Gravitational Effects of Lorentz Violation," Theoretical Physics Group Seminar, University of South Carolina, January 2025.
- "Topics in Beyond the Standard Model Physics," Theoretical Physics Group Seminar, University of South Carolina, September 2024.
- "Aspects of Lorentz Violation: Cause and Effects," Theoretical Physics Group Seminar, University of South Carolina, September 2023.
- Workshop attended (no talk): IUCSS Workshop on Gravitational Aspects of Lorentz Violation, Indiana University, Bloomington, 2023.

Teaching and Outreach

Teaching Assistant, University of South Carolina 2019–2025

- PHYS 201, General Physics I (mechanics), Semesters VIII, IX, X: led weekly discussion sections of about 40 students; wrote and graded homework, quizzes and exam problems.
- PHYS 202, General Physics II (electromagnetism), Semesters II-VIII: assisted with laboratory sessions; prepared tutorial material and Python demonstrations.

Science Outreach Volunteer, IISER Kolkata 2015–2019

Physics demonstration workshops (mechanics, wave optics, astronomy) for local high-school students.

Fellowships and Service

- INSPIRE Fellow, Department of Science and Technology, Government of India, 2014–2019.
- Co-referee, with Prof. B. Altschul, for *Phys. Rev. D.*

Computational Skills

Python, C, MATLAB, Mathematica; numerical solution of ODE and PDE systems; symbolic tensor algebra; LaTeX; Linux.

References

Prof. Brett Altschul
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